

Cartesian diver

Now, let's turn our attention to the Cartesian diver, also known as the Cartesian devil, to better understand the buoyant force.

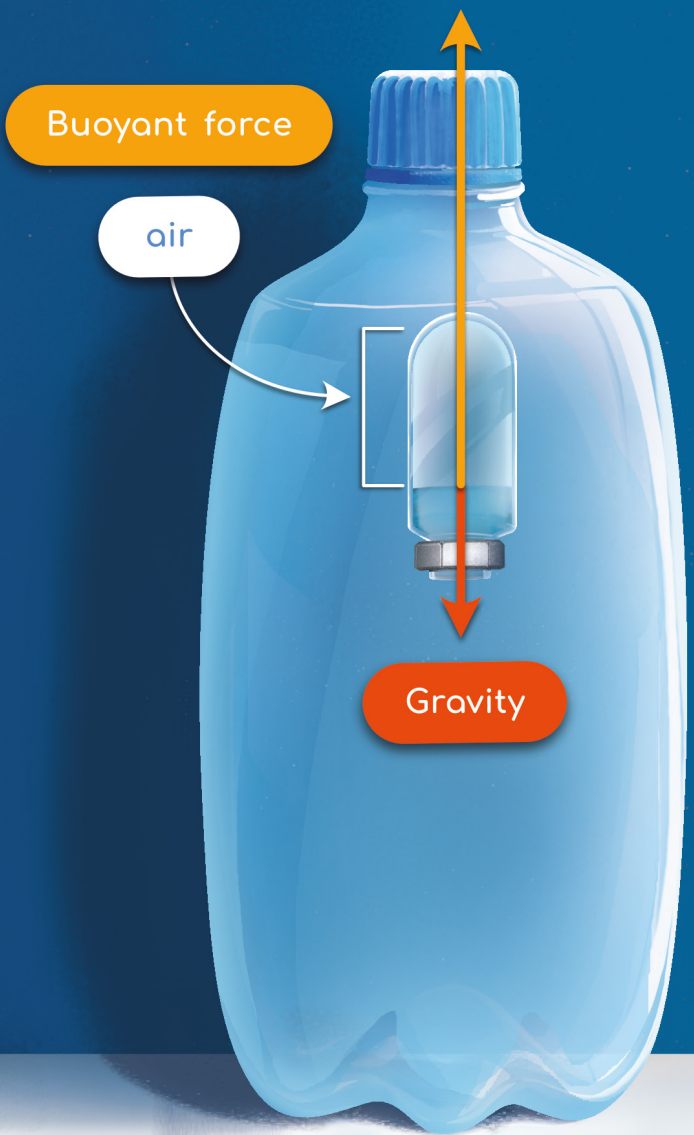
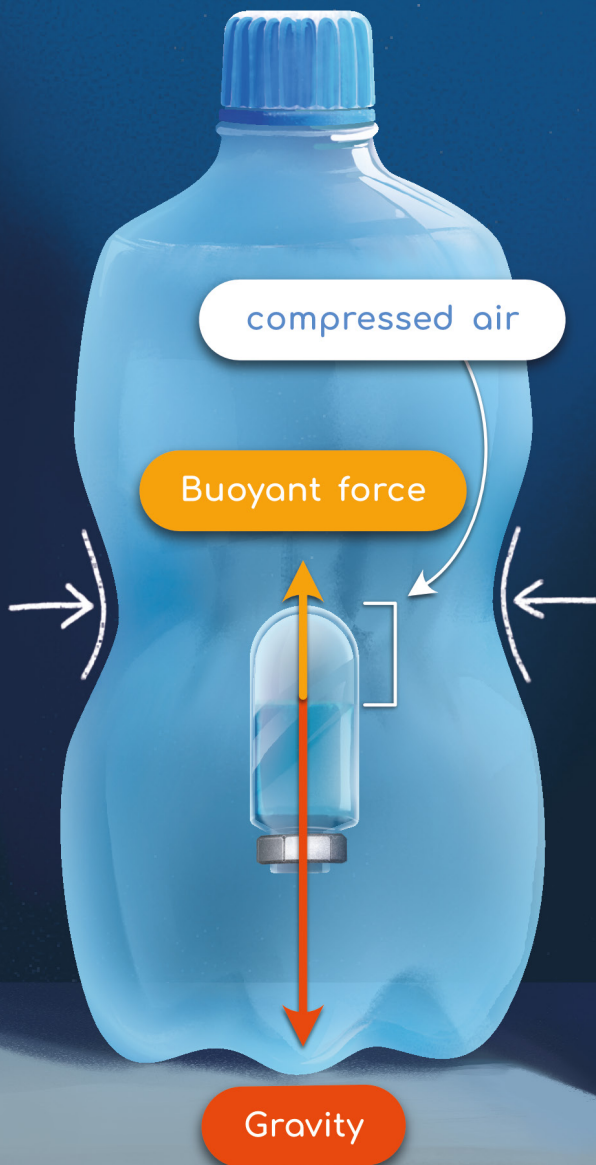
The Cartesian diver was first described in 1648 by Raffaello Magiotti, one of Galileo's students. While no one is quite sure why the toy was named after the French mathematician and philosopher René Descartes, some believe it may be because the toy really makes you think - and as you may know, Descartes is famous for the quotation "cogito, ergo sum" (I think, therefore I am).

The Cartesian diver is made up of a capsule or medicine dropper containing an air-filled bubble which is, itself, enclosed within a bottle of water. The capsule is the diver.

If you place the bottle on a table and avoid disturbing it, you'll notice that the diver will stay at the top of the bottle; however, if you squeeze the bottle, the diver will begin to sink. The harder you squeeze the bottle, the faster the diver will plunge! But as soon as you release the bottle, you'll find that the diver floats to the top once again. Let's explore what causes the Cartesian diver to behave as it does.



There are two forces acting on the Cartesian diver: gravity and the buoyant force. The buoyant force is the strongest, at first, which is why the diver floats at the very top of the bottle. The amount of force depends on the air volume trapped inside the capsule.



When you squeeze the bottle, the water inside is under pressure. But because water is difficult to compress, what actually happens is that air is squeezed into the capsule. As the volume of air inside the capsule decreases, the buoyant force is reduced. At some point it is overcome by the force of gravity acting on the capsule, and this shift causes the capsule to sink.

Interestingly, this principle is used in submarines as well! Submarines have a special compartment known as the ballast tank system, which can be filled with air or water. When filled with air using a pump system, the submarine floats upward. On the other hand, when the tank's vents are opened and water flows in, the submarine will sink. The system can be controlled so well that the submarine's captain can easily achieve zero buoyancy, which means that the submarine neither floats nor sinks. Zero buoyancy happens if the Buoyant force is exactly equal to the gravitational force.

It's easy to find zero buoyancy by playing around with the Cartesian diver. Try it yourself! Can you keep the capsule from rising or sinking?



Ballast tanks

